

Overeducation Persistence Over the Lifecycle: Disentangling Frictions, Specialization and Aging

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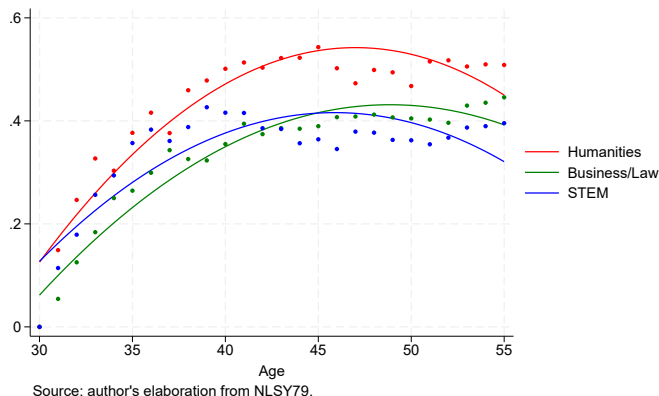
Abstract

I study overeducation persistence with a directed-search model in which workers differ in education, field of study, field-specific experience, and age. Calibrated to the NLSY79 and O*NET, four counterfactual scenarios decompose persistence into its structural channels. Search frictions are by far the dominant driver: removing them would reduce monthly persistence from 97% to 50% early in the career. The specialization trap—non-transferable field-specific experience—is the second most important channel, and learning frictions contribute to a smaller extent. Age effects are negligible. Education is treated as exogenous to focus on post-schooling dynamics. Policies that reduce matching frictions, particularly early in the career, are most effective.

1 Introduction

Why do some workers remain formally overqualified for their job for many years, even decades? Roughly 20–35% of workers globally hold an educational attainment higher than the typical requirement of the job they are employed in (Leuven and Oosterbeek (2011); McGuinness et al. (2018)), we refer to this condition as overeducation. However, most existing explanations treat overeducation as a transient mismatch that resolves as workers gain experience and find better-matched positions. This paper questions that premise. Figure 1 tracks the share of workers employed in a matched job (i.e., a job that typically requires the worker’s education level) among individuals who were overeducated at age 30 in the 1979 National Longitudinal Survey of Youth (NLSY79). While many initially transition into better matches — after five years roughly 30% are no longer overeducated— only 40-45% are employed in a matched job after ten years, a share that barely changes over the following decade.

Figure 1: Share of workers in matched jobs among workers who were overeducated at age 30.



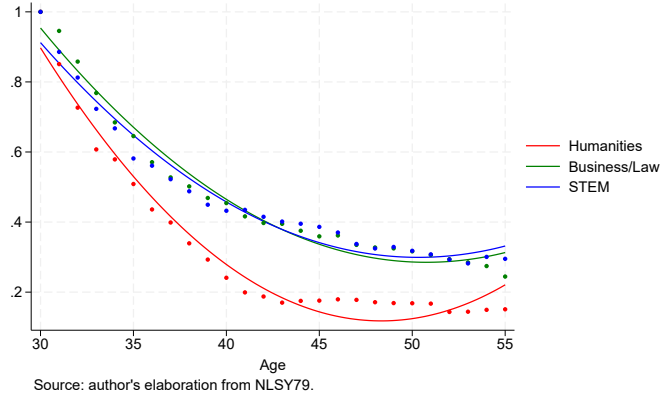
Note: This figure follows all the individuals in the sample who are classified as *overeducated* at age 30 (NLSY79). The x-axis reports the age starting from 30; the y-axis shows the fraction of the original subsample that is employed in a matched job. Overeducation is defined as having attained a higher education level than the occupation’s typical requirement, while matched jobs are those where the worker’s education level matches the occupation’s typical requirement (as specified by O*NET). See Section 2.1 for sample restrictions and further details. See Section 2.2 for details on the overeducation definition.

Moreover, Figure 2 shows that a substantial fraction of workers who were overeducated at age 30 never transition to a matched job in their career. Approximately 30% never find a matched job. This pattern suggests that, at least for some workers, overeducation is not merely a temporary phenomenon, but rather a persistent condition.

To investigate the mechanisms behind this persistence, I develop a directed search model, in the spirit of Menzio et al. (2016) and Menzio and Shi (2011), in which workers differ in educational level, field of study, field-specific experience, and age. Directed search is the natural framework here: it allows workers to optimally target submarkets, trading off the generosity of a contract against the probability of finding a job. This makes the decision to remain in an overeducated position a rational, forward-looking choice rather than something that simply happens to workers. On the firm side, occupations vary in their typical educational requirement and in the productivity of matching workers with different education levels.

I calibrate the framework using U.S. individual-level data from the NLSY79 and information on occupations and educational requirements from O*NET. The model successfully replicates the empirical patterns of overeducation persistence observed in the data without directly targeting them in calibration. I then use the calibrated model to decompose persistence through four counterfactual scenarios. First, removing search frictions is by far the most impactful: early in the career, monthly persistence would fall from 97% to 50–60%, and even after age 45 it would drop from 99% to around 85%. Second, shutting down the specialization trap—by allowing workers to retain

Figure 2: Share of workers who never transitioned to matched jobs among workers who were overeducated at age 30.



Note: This figure follows all the individuals in the sample who are classified as *overeducated* at age 30 (NLSY79). The x-axis reports the age starting from 30; the y-axis shows the fraction of the original subsample that never transitioned to a matched job in their career. Overeducation is defined as having attained a higher education level than the occupation's typical requirement, while matched jobs are those where the worker's education level matches the occupation's typical requirement (as specified by O*NET). See Section 2.1 for sample restrictions and further details. See Section 2.2 for details on the overeducation definition.

field-specific experience when switching fields—would reduce persistence to about 90% when young and 94% later. Third, eliminating learning frictions has a smaller but non-negligible effect, lowering persistence to around 94% early on. Finally, removing aging effects has virtually no impact on persistence throughout the lifecycle.

This paper contributes mainly to two strands of the literature. First, it adds to the body of work on overeducation, dating back to Freeman (1975), who argued that returns to higher education declined in the 1970s. Much of the empirical work that followed, inspired by Duncan and Hoffman (1981), focused on the effects of overeducation on wages and employment (e.g., Verdugo and Verdugo (1989)). Given concerns about omitted variable bias (see Leuven and Oosterbeek (2011)), several studies attempted to address this using instrumental variables (e.g., Korpi and Tåhlin (2009)) or fixed effects (e.g., Bauer (2002); Dolton and Vignoles (2000)). However, the accuracy of these results is often questioned due to the difficulty of finding credible instruments and the possibility that workers sort into occupations based on unobserved characteristics (Leuven and Oosterbeek (2011)). Importantly, most of this literature has emphasized cross-sectional aspects (e.g., Alba-Ramirez (1993); Hartog (2000)), with relatively little attention to dynamics. Notable exceptions include Rubb (2003) and Clark et al. (2017), who provide empirical evidence of substantial persistence in overeducation. This observed persistence is hard to reconcile with existing theories. Human capital theory implies that all the investment in human capital sustained by individuals will eventually be rewarded by firms. The career mobility view (Sicherman and Galor (1990); Sicher-

man (1991)) suggests that workers accept initially mismatched jobs as stepping stones to better-matched positions, implying a temporary phenomenon. In a similar spirit, Dolado et al. (2009) develop a random search model in which workers, endowed with two skill levels and the option to search while employed (unlike in Gautier (2002)), may temporarily accept an overqualified position due to labor market frictions, but eventually transition to a matched job. None of these models, however, can explain the long-term persistence documented in this work. This paper provides the first structural decomposition of overeducation persistence into its constituent channels over the lifecycle.

Second, this paper contributes to the literature on directed search theory, initiated by Montgomery (1991) and further developed by Moen (1997). The model builds closely on Menzio et al. (2016), but introduces additional heterogeneity along two dimensions: workers differ not only in general experience, age, match quality, and employment status, but also in field of study, and field-specific experience. Each dimension does meaningful economic work. The field of study creates asymmetric access to different occupations. The fully non-transferable field-specific experience is the engine of the specialization trap. The ability learning process generates an option value of remaining in the current job that rises with tenure, reinforcing persistence.

The remainder of the paper is organized as follows. Section 2 describes the data and provides descriptive evidence. Section 3 presents the theoretical framework. Section 4 details the calibration, and validates the model. Section 5 presents the decomposition of each channel of overeducation persistence by analyzing four counterfactual scenarios. Section 6 concludes.

2 Data and Descriptive Evidence

This section describes the data and documents the empirical patterns regarding overeducation and its dynamic over the lifecycle. Section 2.1 describes the primary dataset and sample construction. Section 2.2 defines education groups, fields of study, and the overeducation classification. Sections 2.3–2.8 present descriptive evidence on the incidence of overeducation and undereducation, the lifecycle patterns of transitions across employment states, and the age profile of wages by educational field and mismatch.

2.1 Data and Sample

The primary dataset is the National Longitudinal Survey of Youth 1979 (NLSY79), a panel study that has followed a nationally representative sample of 12,686 U.S. citizens born between 1957 and 1964, interviewed annually from 1979 to 1994 and biannually

thereafter. Its longitudinal structure allows me to track individuals' career paths over their working lives, which is essential for studying the long-term dynamics of overeducation.

I complement the NLSY79 with the O*NET database, which contains hundreds of standardized, occupation-specific descriptors for nearly 1,000 occupations covering the entire U.S. economy.

The analysis focuses on prime-age individuals between 30 and 55, covering the period when most workers have completed their formal education and are fully integrated into the labor market. Following standard sample construction (see Baley et al. (2022)), I drop individuals with more than two years of military service, weak labor attachment (more than ten years out of the labor market), workers already employed at the start of the sample period, and those with missing ability scores. The final sample consists of 2,546 individuals, yielding a total of 782,094 monthly observations.

2.2 Measuring Education, Fields, and Overeducation Status

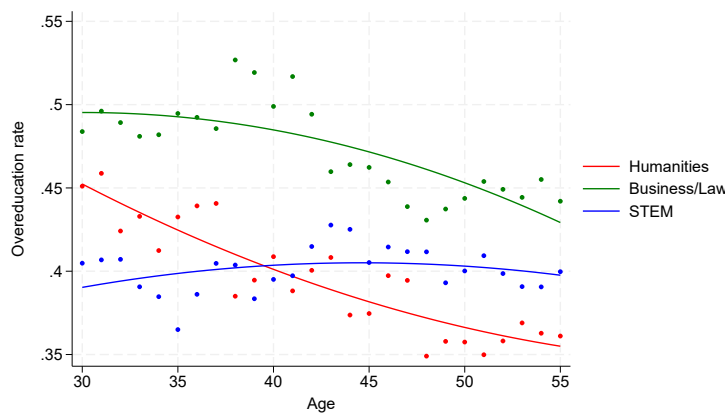
In the empirical part of the paper, individuals are divided into four different educational fields: no college degree if the highest educational attainment is a high school diploma or less ($f_i = 0$), and three college degree fields: $f_i = 1$ for humanities, $f_i = 2$ for business, social sciences or law (business/law hereafter), and $f_i = 3$ for STEM.

Finally, workers are classified as overeducated if they possess a college degree, but they are employed in a job where the typical educational requirement (as specified in the O*NET database) is a high school diploma or less. Conversely, workers are classified as undereducated if they are employed in a job where the typical educational requirement is a college degree, but they do not possess one. Matched workers are those whose educational attainment aligns with the typical requirement of their occupation.

2.3 Incidence of Overeducation

As shown in Figure 3, the incidence of overeducation is substantial, ranging from 50% to 35% across the lifecycle. Graduates in business/law are the most affected, with one worker out of two classified as overeducated at age 30. This share gradually declines to 45% at age 55. A similar pattern is observed for humanities graduates, with a lower incidence of overeducation (45% at age 30 and 35% at age 55). STEM graduates are the least affected when younger, with an incidence of around 40% that remains relatively stable with age. Overall, the incidence of overeducation is high, and its magnitude improves only slightly with age, suggesting that overeducation is not merely a phenomenon affecting young workers.

Figure 3: Share of overeducated workers by age and field.



Source: author's elaboration from NLSY79.

Note: This figure reports the share of college-educated workers classified as overeducated at each age in the NLSY79 sample, separately by field of study (humanities, business/law, and STEM). The x-axis reports worker age (30–55); the y-axis shows the fraction of workers in each field who are overeducated. Overeducation is defined as holding a college degree while being employed in a job whose typical educational requirement is a high school diploma or less (as specified by O*NET). Each series represents the unconditional share within the corresponding field-of-study group. See Section 2.1 for sample restrictions and Section 2.2 for details on the overeducation definition.

2.4 Incidence of Undereducation

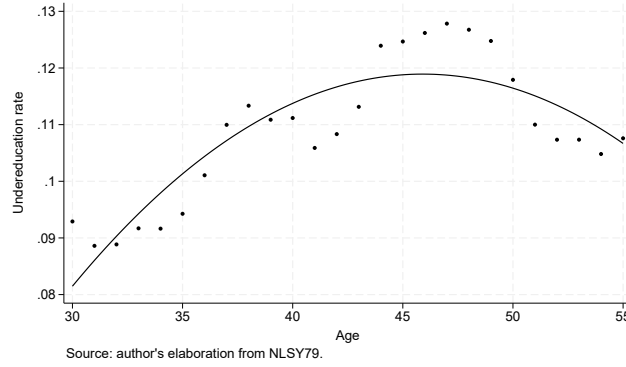
Undereducation follows an opposite pattern, ranging from 9% at the beginning of the career to almost 13% at 45–50 years old (Figure 4), a magnitude consistent with findings in the broader literature (Leuven and Oosterbeek (2011)). This confirms that education alone is not the decisive hiring criterion: other worker characteristics also matter.

More informatively, the share of undereducated workers rises steadily with age. This pattern is consistent with an ability-learning mechanism: as workers accumulate tenure, firms and workers gradually learn the match quality of their current match. When match quality is revealed to be good, lower-educated workers are more likely to stay in college jobs.

2.5 Transitions from Unemployment

Figure 5 documents monthly transition rates out of unemployment for college graduates. The left panel shows the transition rate from unemployment to overeducation, while the right panel shows the transition rate from unemployment to matched employment. Both are relatively stable across the lifecycle, with the transition rate to overeducation fluctuating around 7–10%, while the transition rate to matched employment stays around 10–12%. Importantly, when unemployed many workers enter overeducation, and this share is not declining substantially with age, suggesting that for some workers

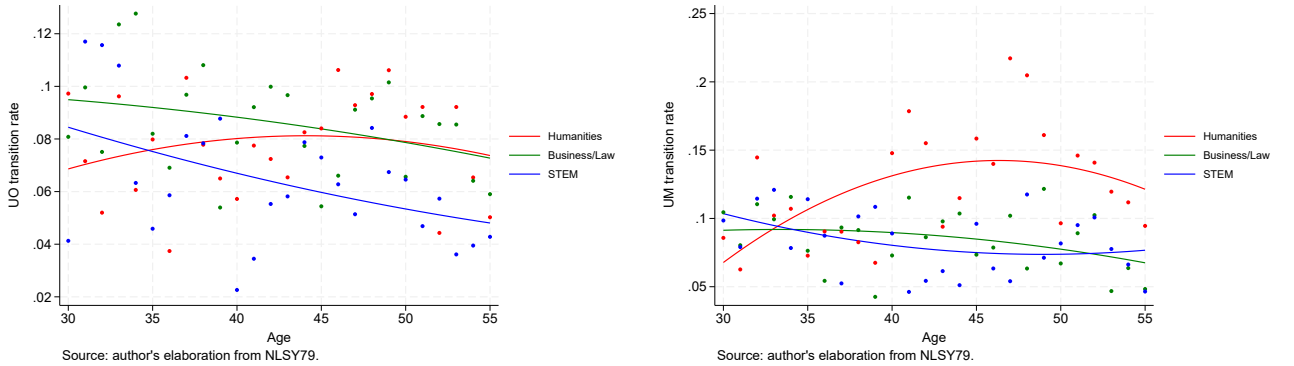
Figure 4: Share of undereducated workers by age.



Note: This figure shows the share of workers classified as undereducated at each age in the NLSY79 sample. The x-axis reports worker age (30–55); the y-axis shows the fraction of all workers who are undereducated. Undereducation is defined as being employed in a job whose typical educational requirement is a college degree (as specified by O*NET) while the worker does not possess one. The series represents the unconditional cohort share across all workers. See Section 2.1 for sample restrictions and Section 2.2 for details on the undereducation definition.

overeducation persists even after a job spell ends.

Figure 5: Monthly transition rates from unemployment to overeducation (left panel) and to matched (right panel) by age and field.



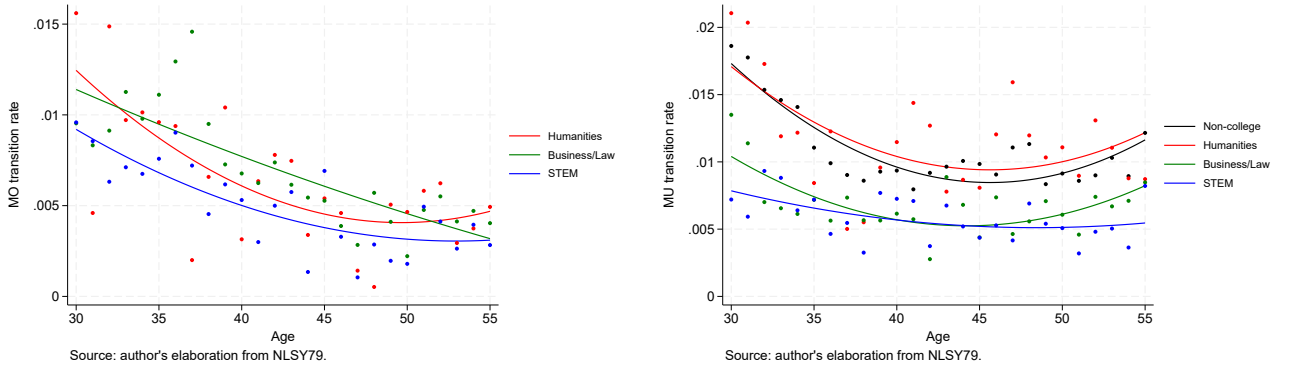
Note: This figure presents monthly transition rates from unemployment to (i) overeducation (left panel) and (ii) matched employment (right panel), by worker age and field. Rates are computed using the NLSY79 sample. Overeducation is defined as having attained a higher education level than the occupation's typical requirement (as specified by O*NET). See Section 2.1 for sample restrictions and Section 2.2 for the overeducation/matched definitions.

2.6 Transitions from Matched

Figure 6 shows monthly transition rates from matched employment to overeducation (left panel) and to unemployment (right panel). The transition rate to overeducation is relatively low, around 0.5-1%, and declines with age. The transition rate to unemployment is higher for non college workers and humanities graduates, ranging from 2%

at the beginning of the lifecycle to approximately 1% at the end. Business/law and STEM graduates are less likely to move to unemployment: 1% when young and 0.5% at the end of their careers. These patterns suggest that in the first half of the lifecycle, matched workers have a non-negligible probability of moving to overeducation or to unemployment (especially non-college workers and humanities graduates), but this probability declines with age, leading to a substantial stability as they get older.

Figure 6: Monthly transition rates from matched to overeducation (left panel) and to unemployment (right panel).



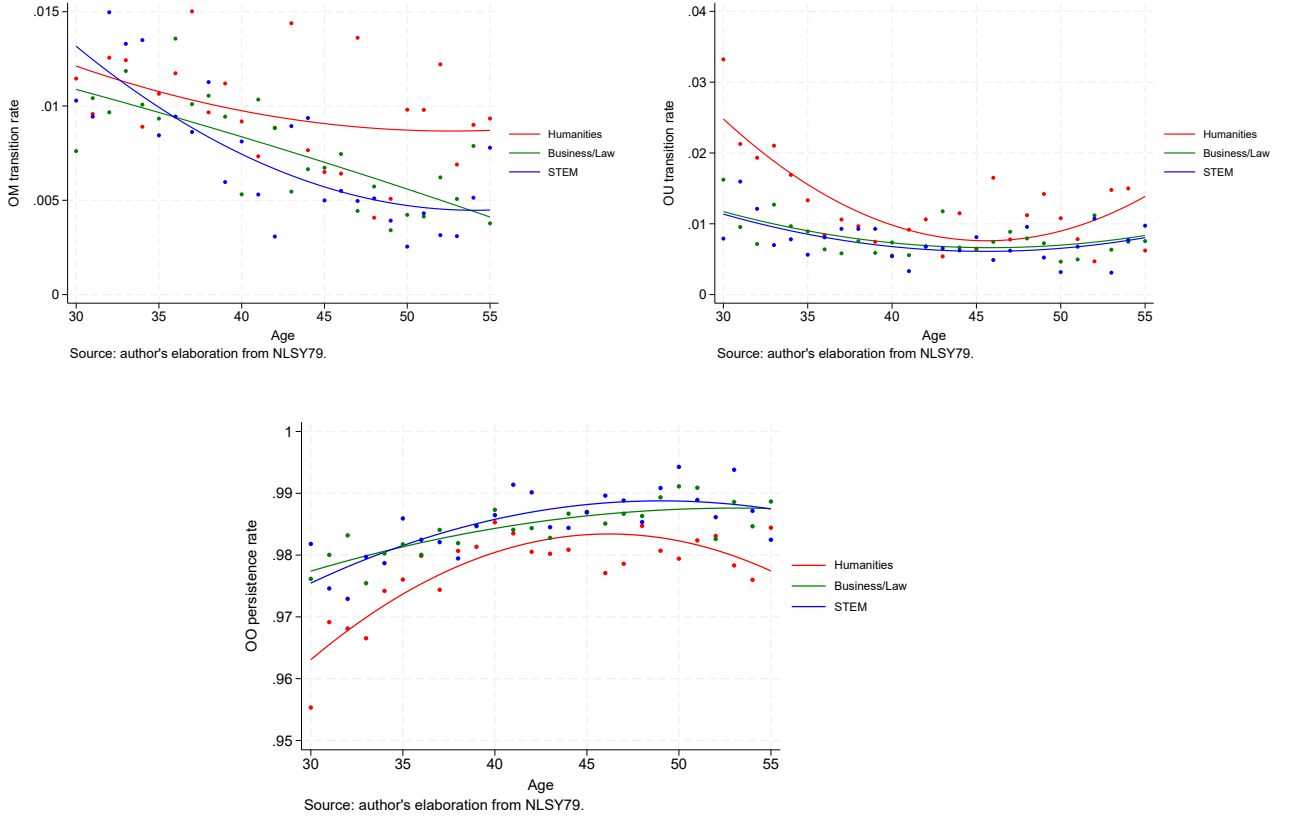
Note: This figure shows monthly transition rates from matched employment to overeducation (left panel) and to unemployment (right panel), separately by field of study (no college, humanities, business/law, and STEM). The x-axis reports worker age (30–55); the y-axis shows the monthly probability of transitioning from a matched job to the indicated state. Rates are computed from the NLSY79 sample. A worker is classified as matched when her educational attainment aligns with the typical requirement of her occupation (as specified by O*NET), and as overeducated when she holds a college degree but is employed in a job requiring only a high school diploma or less. See Section 2.1 for sample restrictions and Section 2.2 for the overeducation/matched definition.

2.7 Transitions from Overeducation and Persistence

In the top-left panel of Figure 7, the transition rate from overeducation to matched employment is shown. This is the monthly share of overeducated workers that manage to exit overeducation and find a matched position. The transition rate is higher when workers are younger, around 1-1.5% for all fields, but declines with age, stabilizing around 0.5% in the second half of the working life. The top-right panel shows the transition rate from overeducation to unemployment, which is decreasing for humanities graduates, from 2% at the beginning of the lifecycle to 1% at the end, while it is relatively stable for business and STEM graduates, around 1% for their whole lifecycle. Finally, the bottom panel shows the persistence rate, that is, the monthly share of overeducated workers that remain overeducated. The persistence rate is quite high: around 97% in the first half of the lifecycle and reach 99% in the second half. This

means that on average in a given year, out of 100 overeducated workers, 70 remain overeducated when young and 89 when older. Importantly, the fact that overeducation persistence increases with age suggests that, as workers get older, some mechanisms make it more difficult to exit overeducation.

Figure 7: Monthly transition rates from overeducation to: matched (left panel), unemployment (right panel), and remaining overeducated (bottom panel).



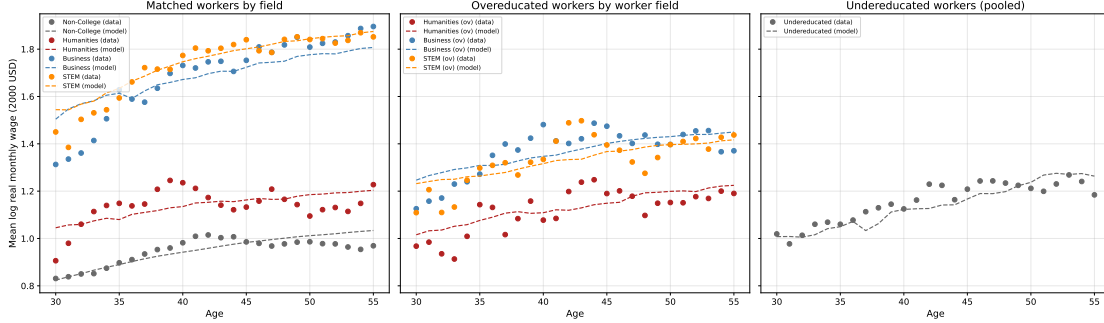
Note: This figure presents monthly transition rates from overeducation to (i) matched (OM rate), (ii) unemployment (OU rate), and (iii) remaining overeducated (Persistence rate). Rates are computed using the NLSY79 sample. Overeducation is defined as having attained a higher education level than the occupation's typical requirement (as specified by O*NET). See Section 2.1 for sample restrictions and Section 2.2 for the overeducation/matched definition.

2.8 Age Profile of Wages

This section presents the age profile of wages by educational field and mismatch status. Figure 8 shows the average monthly wage for matched, overeducated, and undereducated workers in each field. Matched workers (left panel) consistently earn higher wages than their overeducated counterparts (center panel) across all fields, with business/law and STEM graduates earning the most. Humanities graduates earn more than

non-college workers but significantly less than the other college graduates. Undereducated workers (right panel) suffer a penalty as they earn less than matched workers in college jobs, but their wage is higher than non-college matched workers. These patterns are particularly relevant for the model calibration, as they provide information on the productivity of workers in different education-job matches and the wage penalties associated with overeducation and undereducation.

Figure 8: Age Profile of Wages



Note: Average log monthly wages are plotted by age and education-job match status for each field. The left panel shows matched workers, the middle panel shows overeducated workers, and the right panel shows undereducated workers. Wages are computed using the NLSY79 sample and are expressed in real monthly terms. The figure highlights the wage premium of matched work and the wage penalties associated with overeducation and undereducation across fields. Dots represent actual data from the NLSY79 sample, while lines show wages implied by the calibrated model. Overeducation is defined as having attained a higher education level than the occupation's typical requirement (as specified by O*NET). See Section 2.1 for sample restrictions and Section 2.2 for the overeducation/matched definition.

3 Theoretical Framework

3.1 The Environment

The economy consists of T overlapping generations. Time is discrete, and $t \in \{0, 1, \dots, T\}$ denotes the age of the worker. Workers and firms are risk-neutral. Each worker is endowed with one indivisible unit of labor and maximizes the expected discounted sum of per-period consumption with discount factor $\beta \in (0, 1)$. A continuum of firms, with positive measure, populate the economy.

Each occupation is associated with a field $f_j \in \{0, 1, \dots, F\}$. If $f_j = 0$, the occupation is not associated with any college major and therefore does not require a college degree. If $f_j > 0$, the occupation generally requires a college degree in field f_j . Workers' experience $e_{if} \in \{0, 1, \dots, t\}$ corresponds to the number of past periods worked in field f . Importantly, experience is fully non-transferable across fields, so accumulated experience in non-college jobs is not productive for college jobs, and vice versa.

The output of a firm–worker match is

$$y(a_{ij}, f_i, f_j, e_{if}, t) = \begin{cases} g(f_i, f_j, e_{if}, t) & \text{if } a_{ij} = a^G \\ 0 & \text{if } a_{ij} = a^B \end{cases} \quad (1)$$

where a_{ij} denotes the current firm–worker match quality, that is initially unobserved by both parties. Each period while employed, with probability α , the true value of a_{ij} is revealed; until then it is denoted as $a_{ij} = a^0$. Moreover, the true value of $a_{ij} \in \{a^B, a^G\}$ can take only two values, good or bad. If the match is of good quality, the productivity of the match depend on the worker’s education field f_i and the job’s field f_j , on the worker’s accumulated experience in the field of the job, e_{if} , and on the age t of the worker, to capture possible losses in productivity due to aging. If instead match quality is bad, the worker is not productive and the match produces zero output. The production function $g(f_i, f_j, e_{if}, t)$ is given by

$$g(f_i, f_j, e_{if}, t) = [\rho_f + \text{mismatch}(f_i, f_j)] h(e_{if}, t)$$

where ρ_f denotes the baseline productivity of a matched worker ($f_i = f_j$), $\text{mismatch}(f_i, f_j)$ is the productivity premium (penalty) of an overeducated (undereducated) worker relative to a matched worker, and $h(e_{if}, t)$ is an increasing function of field specific experience and a decreasing function of age.

3.2 Education and Fields

Since this paper focuses on post-schooling labor market dynamics, education is treated as predetermined at labor market entry. Each worker is endowed with an education field $f_i \in \{0, 1, \dots, F\}$, where $f_i = 0$ denotes no college degree and $f_i > 0$ denotes a college degree in field f_i .

When match quality is unknown, agents form expectations based on the worker’s education field f_i and the job’s field f_j . In particular, the probability of good match quality is

$$P(a_{ij} = a^G | f_i, f_j) = \begin{cases} \bar{a}_f^{\text{baseline}} & \text{if } f_j = 0 \text{ or } f_i = f_j > 0 \\ \bar{a}^{\text{mismatched}} & \text{if } f_i \neq f_j > 0 \end{cases}$$

where $\bar{a}_f^{\text{baseline}}$ is the probability of a good match quality for matched workers ($f_i = f_j$). Notice that this probability depends on the field f and for jobs that do not require a college degree the probability of a good match quality is the same both for overeducated and no-college workers. On the other hand, $\bar{a}^{\text{mismatched}}$ is the probability of a good match

quality for mismatched workers: workers without a college degree ($f_i = 0$) employed in a job that requires a college degree ($f_j > 0$) or workers with a college degree in a field $f_i \neq f_j > 0$.

3.3 The Labor Market

The type of a worker is summarized by $(a_{ij}, f_i, f_j, e_{if}, t) = (\omega, t)$, that is, the education of the worker, f_i ($f_i = 0$ for no college graduate), the education field typically required by her current job, f_j , experience in current job field f_j , e_{if} , current match quality, a_{ij} , and age t . When unemployed, f_j and e_{if} refers to the most recent job and a_{ij} is set to a^0 . Each vacancy must specify (ω, x, t) , where x is the value offered to the worker in lifetime utility. Search is directed, meaning that there is a continuum of submarkets indexed by (ω, x, t) . Workers decide optimally where to search based on their type, and firms decide how many vacancies to create and where to locate them. $\theta_t(\omega, x)$, the ratio between the vacancies and the individuals searching in a certain submarket, is the market tightness of the submarket (ω, x, t) . Workers and vacancies in the same submarket meet through a frictional process. Searching on-the-job is also possible: while employed workers are allowed to search for a job in a different field, but at a lower intensity than unemployed workers. Importantly, when a worker accept a job in another field there is a switching cost to be paid; this can be interpreted as an initial retraining, an additional friction or as a psychological cost.

3.4 Timing, State and Contracts

The rest of the model follows Menzio et al. (2016), adapted to accommodate the heterogeneity described above.

Aggregate state. At the beginning of each period, the aggregate state is $\psi = (n, u, e, \gamma)$, where $n(\omega, t)$ is the measure of non-participants, $u(\omega, t)$ the measure of unemployed, $e(\omega, t)$ the measure of employed, and γ the current realization of the cohort-size process.

Within-period timing. Each period has five stages: entry-and-exit, separation, search, matching, and production.

Entry and exit. A non-participating worker may enter the labor market. Participating workers may exit permanently.

Separation. An employed worker becomes unemployed with probability $d \in [\delta, 1]$, where $\delta \in [0, 1]$ is the exogenous destruction rate and d is set by the employment contract.

Search. Unemployed workers search with probability $\lambda_u \in (0, 1]$; employed workers who retained their job search on-the-job with probability $\lambda_e \in (0, 1]$; workers who just lost their job cannot search in the current period. When searching, a worker optimally selects a submarket. Firms choose how many vacancies to post in each submarket at a flow cost $k_f > 0$ per vacancy. Importantly, the cost of maintaining a vacancy, k_f , is potentially different across fields. This reflects the different costs and frictions associated with recruiting workers in different positions.

Matching. In submarket (ω, x, t) , a searching worker meets a vacancy with probability $p(\theta_t(\omega, x))$, where $p : \mathbb{R}_+ \rightarrow [0, 1]$ is twice-differentiable, strictly increasing, strictly concave, with $p(0) = 0$ and $p(\infty) = 1$. A vacancy meets a worker with probability $q(\theta) = p(\theta)/\theta$, $q(0) = 1$, $q(\infty) = 0$. Upon meeting, the firm offers a contract worth x in lifetime utility; the worker accepts or returns to her previous status.

Production. An unemployed worker produces and consumes $b(a^H, f_i, f_j, e_{if}, t)$ (the unemployment flow value depends on the worker's characteristics and last job). An employed worker produces $y(a_{ij}, f_i, f_j, e_{if}, t)$ and receives wage w as specified in her contract. Match quality is revealed with probability $\alpha \in (0, 1]$. At the end of the period, nature draws next period's cohort size from $\Pi(\hat{\gamma}|\gamma)$. Hats denote next-period variables throughout.

Contracts. Employment contracts are complete: they specify the wage w , the endogenous separation probability d , and the submarket where the employee should search, all as functions of the match history and the aggregate state. Firms therefore offer bilaterally efficient contracts that maximize the joint surplus of the match (see Menzio and Shi (2011)).

3.5 Characterization of the Equilibrium

The value of unemployment for a worker of type $(\omega, t) = (a_{ij}, f_i, f_j, e_{if}, t)$ is

$$U_t(\omega, \psi) = b(a^H, f_i, f_j, e_{if}, t) + \beta \mathbb{E}_{\hat{\psi}|\psi} [U_{t+1}(\omega, \hat{\psi}) + \lambda_u R_{t+1}(\omega, \hat{\psi})]$$

$$R_{t+1}(\omega, \hat{\psi}) = \max_{x, f'_j} \{p(\theta_{t+1}(\omega', x, \hat{\psi}))[x - U_{t+1}(\omega, \hat{\psi})]\}$$

In the current period, the unemployed produces and enjoys b : the unemployment flow value increases with the worker's productivity in the most recent job. In the following period, with probability λ_u the worker can search into the labor market, where she can choose to search for a job in a certain field f'_j . The optimal submarket is chosen by maximizing the probability of finding a job, $p(\theta)$ (decreasing in x), times the surplus deriving from the new employment condition, $x - U_{t+1}$.

The sum of the firm's lifetime profits and the worker's lifetime utility in a match with known quality $a \neq a^0$ is

$$\begin{aligned} V_t(\omega(a), \psi) = & y(a_{ij}, f_i, f_j, e_{if}, t) + \beta \mathbb{E}_{\hat{\psi}|\psi} \left[\max_{d \in [\delta, 1]} \{ dU_{t+1}(\hat{\omega}(a), \hat{\psi}) + \right. \\ & \left. + (1-d)(V_{t+1}(\hat{\omega}(a), \hat{\psi}) + \lambda_e S_{t+1}(\hat{\omega}(a), \hat{\psi})) \} \right] \\ S_{t+1}(\hat{\omega}(a), \hat{\psi}) = & \max_{x, f'_j} \{ p(\theta_{t+1}(\omega', x, \hat{\psi})) [x - V_{t+1}(\hat{\omega}(a), \hat{\psi})] \} \end{aligned}$$

where $\hat{e}_{if} = e_{if} + 1$. In the current period, the employed produces y , which depends on the match quality, a_{ij} , the field of the worker and the job, f_i and f_j , the accumulated experience in field f_j , e_{if} , the age of the worker, t . In the following period, with probability d the match is destroyed and the worker moves to unemployment. Otherwise, the match continues, but the possibility of search on-the-job must be taken into account. In particular, an employed worker has the opportunity to search for a job in a different field with probability λ_e . In this case, the worker optimally chooses the submarket where to search.

The sum of the firm's lifetime profits and the worker's lifetime utility in a match with unknown ability $a = a^0$ is

$$\begin{aligned} V_t(\omega(a^0), \psi) = & \alpha \mathbb{E}[V_t(\omega(a), \psi)] + (1-\alpha) \mathbb{E}[y(a_{ij}, f_i, f_j, e_{if}, t)] + \\ & + (1-\alpha) \beta \mathbb{E}_{\hat{\psi}|\psi} \max_{d \in [\delta, 1]} \{ dU_{t+1}(\hat{\omega}, \hat{\psi}) + \\ & + (1-d)(V_{t+1}(\hat{\omega}(a^0), \hat{\psi}) + \lambda_e S_{t+1}(\hat{\omega}(a^0), \hat{\psi})) \} \end{aligned}$$

In the current period, with probability α agents discover the true match quality, a_{ij} , otherwise they have an expectation of the output. The following period is equivalent to the previous value function. The two expectation terms are given by:

$$\begin{aligned} \mathbb{E}[V_t(\omega(a), \psi)] &= P(a_{ij} = a^G) [V_t(\omega(a^G), \psi)] + (1 - P(a_{ij} = a^G)) [V_t(\omega(a^B), \psi)] \\ \mathbb{E}[y(a_{ij}, f_i, f_j, e_{if}, t)] &= P(a_{ij} = a^G) [y(a^G, f_i, f_j, e_{if}, t)] + (1 - P(a_{ij} = a^G)) [y(a^B, f_i, f_j, e_{if}, t)] \end{aligned}$$

where

$$P(a_{ij} = a^G) = \begin{cases} \bar{a}_f^{baseline} & \text{if } f_j = 0 \text{ or } f_i = f_j > 0 \\ \bar{a}^{mismatched} & \text{if } f_i \neq f_j > 0 \end{cases}$$

is the probability that match quality is high, which depends on the educational (mis)match.

When a firm and a worker meet in a submarket targeting field f'_j —whether the worker is currently unemployed or searching on the job—they form a new match with

unknown quality a^0 . The new match state is $\omega' = (a^0, f_i, f'_j, e', t)$, where accumulated experience e' is retained ($e' = e_{if}$) if $f'_j = f_j$, but resets to zero ($e' = 0$) if the worker switches to a different field. Furthermore, if the worker switches fields, a field-specific proportional switching cost c_{s,f_i} is paid. This cost reduces the effective total value of the new match to $\tilde{V}_{t+1}(\omega', \hat{\psi})$, defined as:

$$\tilde{V}_{t+1}(\omega', \hat{\psi}) = V_{t+1}(\omega', \hat{\psi}) \times \begin{cases} 1 & \text{if } f'_j = f_j \\ 1 - c_{s,f_i} & \text{if } f'_j \neq f_j \end{cases} \quad (2)$$

The zero-profit condition for firms posting a vacancy in submarket (ω', x, t) implies that the firm's expected surplus must equal the field-specific vacancy cost $k_{f'_j}$:

$$k_{f'_j} \geq q(\theta_{t+1}(\omega', x, \hat{\psi})) \left[\tilde{V}_{t+1}(\omega', \hat{\psi}) - x \right] \quad (3)$$

and $\theta_{t+1}(\omega', x, \hat{\psi}) \geq 0$ with complementary slackness. This condition ensures that firms behave optimally since, in submarkets with at least a vacancy, the cost of maintaining open a vacancy, k_f , must equal the expected benefit of an open vacancy (RHS), that is the probability of meeting a worker, $q(\theta)$, times the surplus extracted by the firm, $V - x$. If instead a submarket is empty ($\theta = 0$), it must be that the cost of the vacancy is greater than the expected benefit for the firm.

The unique equilibrium is block-recursive (BRE): a recursive equilibrium in which the agents' value and policy functions do not depend on the aggregate state of the economy ψ . The proof is reported in Appendix A and follows Menzio et al. (2016) closely.

4 Calibration and Validation

4.1 Calibration

Production function and functional forms. I specify the production function as follows. For a worker with education field f_i employed in a job of field f_j , with field-specific experience e_{if} and age t , output when the match is good ($a_{ij} = a^G$) is

$$g(f_i, f_j, e_{if}, t) = [\rho_{f_j} + \text{mismatch}(f_i, f_j)] h(e_{if}, t),$$

The baseline productivity ρ_{f_j} is a field-specific parameter: ρ_0 for non-college occupations ($f_j = 0$), and ρ_{f_j} for college occupations in field $f_j \in \{1, 2, 3\}$ (humanities, business, STEM, respectively).

The mismatch term is specified as

$$\text{mismatch}(f_i, f_j) = \begin{cases} 0 & \text{if } f_i = f_j \quad (\text{matched}), \\ \mu^{ov}(f_i) & \text{if } f_i > 0, f_j = 0 \quad (\text{overeducated}), \\ \mu^{mm} & \text{if } f_i \neq f_j, f_j > 0 \quad (\text{mismatched}), \end{cases}$$

where $\mu^{ov}(f_i)$ captures the productivity premium of a college graduate in field f_i employed in a non-college job (notice that the premium varies across the worker's field of study), and μ^{mm} captures the penalty of a worker whose field does not match the job's field requirement.

The human-capital component is

$$h(e_{if}, t) = [(1 - \phi_{1,f_j}) + \phi_{1,f_j}(1 + e_{if})^{\phi_2}] - \phi_3 t,$$

where $\phi_{1,f_j} > 0$ is a field-specific parameter that governs the weight of accumulated experience in field f_j , $\phi_2 \in (0, 1)$ determines the degree of concavity of experience returns (shared across fields), and $\phi_3 \geq 0$ captures productivity losses due to skill depreciation and aging. The field-specific ϕ_{1,f_j} allows experience returns to differ across occupational fields, reflecting the fact that the value of accumulated know-how may vary substantially between, say, non-college and STEM jobs.

Following Menzio et al. (2016), the matching probability takes the form $p(\theta) = \min\{\theta^{1/2}, 1\}$.

Free parameters. The model has 24 free parameters, calibrated in two steps: 14 production-function parameters are calibrated to match lifecycle wage profiles, and 10 search-and-matching parameters are calibrated to match age-specific labor market transition rates. An additional 8 parameters are either normalized or assigned from external sources. The 14 production-function parameters are: the four baseline field productivities $\rho_0, \rho_1, \rho_2, \rho_3$; the three overeducation premia $\mu^{ov}(1), \mu^{ov}(2), \mu^{ov}(3)$; the undereducation penalty μ^{mm} ; the four field-specific experience weights $\phi_{1,0}, \phi_{1,1}, \phi_{1,2}, \phi_{1,3}$; the concavity ϕ_2 ; and the depreciation rate ϕ_3 . The 10 search-and-matching parameters are: the exogenous separation rate δ ; the employed search intensity λ_e ; the four field-specific vacancy costs k_0, k_1, k_2, k_3 ; the three field-specific switching costs $c_{s,1}, c_{s,2}, c_{s,3}$; and the probability of a good match for mismatched workers $\bar{a}^{mismatched}$.

Step 1: Production function (wages). The 14 production-function parameters are calibrated by minimizing the distance between model-implied average log wages and their empirical counterparts from Section 2.8. I assume wages are a constant fraction

of match productivity.¹ For each education-job-field cell and age, the model predicts log wages as $\log(w) = \log(\rho_{f_j} + \text{mismatch}(f_i, f_j)) + \log(h(e_{if}, t))$, using the empirical distribution of field-specific experience (partitioned into deciles from the NLSY79). Figure 8 confirms that the calibrated model replicates the lifecycle wage profiles across all education-field cells with reasonable accuracy: the model matches the level differences across fields and the concave age profile of wages for matched, overeducated, and undereducated workers.

Step 2: Assigned and normalized parameters. Several parameters are assigned directly from external sources or normalization. λ_u is normalized to 1, and the switching cost for non-college workers is normalized to $c_{s,0} = 0$. The discount factor β is set so that $\beta^{-12} - 1 = 4.04\%$, the average Federal Funds Rate between 1982 and 2020. The ability learning probability $\alpha = 0.264$ is taken from Menzio et al. (2016), who estimate it using a similar directed search framework. The unemployment flow value is specified as $b = 0.71 \times y(a^G, f_i, f_j, e_{if}, t)$, so that it equals 71% of the worker’s potential output in her most recent job, following the wage-replacement ratio estimated by Hall and Milgrom (2008). The field-specific probabilities of a good match for matched workers, $\bar{a}_f^{\text{baseline}}$, are assigned from the NLSY79 data: for each job field f , I compute the share of new, correctly matched hires that do not end in separation within 12 months. The reasoning is that, after one year, $\alpha = 0.264$ implies there is a high probability (more than 97%) that match quality has been revealed, so the share of new hires who remain employed in the same job after one year is a good proxy for the probability of a good match.

Step 3: Search and matching parameters (targeted moments). The 10 remaining free parameters— δ , λ_e , k_0, k_1, k_2, k_3 , $c_{s,1}, c_{s,2}, c_{s,3}$, and $\bar{a}^{\text{mismatched}}$ —are calibrated by targeting the age-specific monthly transition rates observed in the NLSY79 for workers aged 30–55. For each of the 26 ages in this range, I target three aggregate rates: the unemployment-to-employment rate (UE), the employment-to-unemployment rate (EU), and the cross-field employment-to-employment rate (EE), for a total of 78 moments.

Model-implied transition rates are computed analytically at each age using the calibrated policy functions and the empirical distribution of worker types from the NLSY79. The UE rate at age t is $\text{UE}_t = \lambda_u p(\theta_u^*(\omega, t))$, where θ_u^* is the optimal tightness chosen by the unemployed worker. The EU rate is $\text{EU}_t = d^*(\omega, t)$, the endogenous separation probability. The EE rate is $\text{EE}_t = (1 - d^*) \lambda_e p(\theta_e^*(\omega, t))$. In all three cases, the aggregate rate at each age is the weighted average across all worker types present at that age

¹This assumption is consistent with the wage-posting structure of the directed search equilibrium.

in the data.

The identification of each parameter group is as follows. The vacancy costs k_{f_j} determine market tightness and thereby primarily affect the UE rate: higher costs reduce vacancy creation and lower matching probabilities. The exogenous separation rate δ is the floor of the endogenous separation policy and directly governs the EU rate. The employed search intensity λ_e and the field-specific switching costs c_{s,f_i} jointly govern the EE rate: λ_e scales the overall intensity of on-the-job search, while c_{s,f_i} shapes the field-specific incentives to switch. Finally, $\bar{a}^{mismatched}$ governs the probability of a productive match for workers employed outside their field and therefore influences the relative attractiveness of cross-field moves.

Table 1 displays the average targeted moments across all ages and confirms that the calibrated model closely matches the data for all three rates. Figure 9 shows the age-specific fit: the model captures the declining UE rate, the falling EU rate, and the relatively stable EE rate observed over the lifecycle.

Table 1: Model vs. Data: average targeted moments (ages 30–55).

Moment	Model	Data	Source
UE rate	0.1480	0.1475	NLSY79
EU rate	0.0097	0.0094	NLSY79
EE rate	0.0033	0.0028	NLSY79

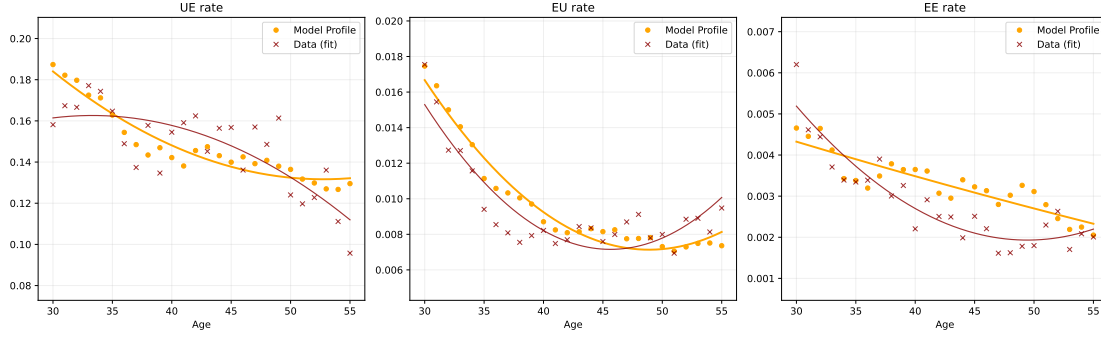
Note: This table reports the average (over ages 30–55) of the three targeted transition rates. “UE rate” is the monthly unemployment-to-employment rate; “EU rate” is the monthly employment-to-unemployment rate; “EE rate” is the monthly cross-field employment-to-employment rate. Data moments are computed from the NLSY79 with sample restrictions detailed in Section 2.1. Model moments are the weighted average of the theoretical transition probabilities across all worker types present at each age in the data. See Figure 9 for the age-specific comparison.

Discussion of calibrated parameter values. Table 2 reports all calibrated and assigned parameters. Several findings deserve discussion.

The production-function parameters reveal substantial heterogeneity across fields. Baseline productivity for Humanities ($\rho_1 = 1.77$), business/law ($\rho_2 = 1.07$), and STEM ($\rho_3 = 1.30$) jobs is significantly higher than non-college jobs ($\rho_0 = 0.80$), rationalizing the college wage premium. All three college fields carry a positive overeducation premium ($\mu^{ov} \in [0.22, 0.47]$), meaning that overeducated college graduates still command higher wages than non-college matched workers in the same type of job. The field-specific experience weights ϕ_{1,f_j} vary markedly: experience in business ($\phi_{1,2} = 11.17$) and STEM ($\phi_{1,3} = 8.82$) jobs carries the highest returns, while humanities jobs ($\phi_{1,1} = 2.05$) exhibit substantially lower returns.

The learning probability $\alpha = 0.264$, taken from Menzio et al. (2016), implies that

Figure 9: Age-specific transition rates: Model vs. Data.



Note: This figure compares age-specific transition rates from the calibrated model and the NLSY79 data (ages 30–55). Three panels show: (i) the unemployment-to-employment (UE) rate, i.e. the monthly probability of an unemployed worker finding a job; (ii) the employment-to-unemployment (EU) rate, i.e. the monthly probability of losing a job; and (iii) the cross-field employment-to-employment (EE) rate, i.e. the monthly probability of switching to a job in a different occupational field. The x-axis reports worker age; the y-axis shows the monthly transition probability. Dots represent raw age-specific rates from the NLSY79; lines are polynomial fits to the model-implied rates. All three rates are directly targeted in the calibration. See Table 1 for the average comparison and Section 2.1 for sample restrictions.

in any given month there is a 26.4% chance that the true match quality is revealed. After 3 months, ability remains unknown in approximately 39.8% of matches; after 6 months, this share drops to 15.8%, and after a year, only 2.5% of matches still have unknown quality. Learning frictions therefore play a significant role early in a job spell, reinforcing the option value of staying in the current job.

The vacancy costs display substantial heterogeneity across fields: $k_0 = 35.90$ for non-college jobs, compared with $k_1 = 22.63$ (humanities), $k_2 = 169.48$ (business), and $k_3 = 164.00$ (STEM). The much higher costs for business and STEM vacancies reflect the greater difficulty of filling specialized positions, which tightens those submarkets.

The field-specific switching costs c_{s,f_i} reveal meaningful differences in labor market flexibility across education types. Business/law ($c_{s,2} = 0.23$) and STEM ($c_{s,3} = 0.28$) graduates face substantial switching costs, equivalent to forfeiting 23–28% of the total match value upon changing fields. This means that there are some forces outside the model (e.g., retraining cost, psychological barriers) that prevents workers from easily switching fields. In contrast, the small negative value $c_{s,1} = -0.02$ for humanities, implies that switching fields is actually slightly beneficial for humanities graduates.

The calibrated probability of a good match for mismatched workers is extremely low, $\bar{a}^{mismatched} = 0.006$, implying that a worker employed outside her field has a very low chance of being productive. In contrast, the field-specific baseline probabilities range from $\bar{a}_0^{baseline} = 0.765$ for non-college jobs to $\bar{a}_3^{baseline} = 0.884$ for STEM jobs.

The calibrated exogenous separation rate $\delta = 0.006$ is low, implying that most separations are endogenous, arising from the revelation of bad match quality or from

optimal on-the-job search. The employed search intensity $\lambda_e = 0.713$ indicates that employed workers search at approximately 71% of the intensity of unemployed workers.

Table 2: Calibrated parameters.

Parameter	Description	Value	Methodology
<i>Panel A: Assigned and normalized</i>			
λ_u	Search intensity when unemployed	1.000	Normalized
$c_{s,0}$	Switching cost: non-college workers	0.000	Normalized
β	Discount factor	0.997	Avg. Federal Funds Rate, assigned
α	Ability learning probability	0.264	Menzio et al. (2016), assigned
$\bar{a}_0^{baseline}$	Prob. good match: non-college jobs	0.765	NLSY79, assigned
$\bar{a}_1^{baseline}$	Prob. good match: humanities jobs	0.681	NLSY79, assigned
$\bar{a}_2^{baseline}$	Prob. good match: business jobs	0.871	NLSY79, assigned
$\bar{a}_3^{baseline}$	Prob. good match: STEM jobs	0.884	NLSY79, assigned
<i>Panel B: Production function (Step 1, calibrated from wage profiles)</i>			
ρ_0	Baseline productivity: non-college	0.796	Wage calibration
ρ_1	Baseline productivity: humanities	1.772	Wage calibration
ρ_2	Baseline productivity: business	1.065	Wage calibration
ρ_3	Baseline productivity: STEM	1.303	Wage calibration
$\mu^{ov}(1)$	Overeducation premium: humanities	0.221	Wage calibration
$\mu^{ov}(2)$	Overeducation premium: business	0.467	Wage calibration
$\mu^{ov}(3)$	Overeducation premium: STEM	0.425	Wage calibration
μ^{mm}	Mismatch penalty	-0.445	Wage calibration
$\phi_{1,0}$	Experience weight: non-college jobs	5.290	Wage calibration
$\phi_{1,1}$	Experience weight: humanities jobs	2.047	Wage calibration
$\phi_{1,2}$	Experience weight: business jobs	11.175	Wage calibration
$\phi_{1,3}$	Experience weight: STEM jobs	8.824	Wage calibration
ϕ_2	Experience concavity	0.068	Wage calibration
ϕ_3	Skill depreciation per month	0.00005	Wage calibration
<i>Panel C: Search and matching (Step 3, calibrated from transition rates)</i>			
δ	Exogenous separation rate	0.006	Transition rates
λ_e	Search intensity while employed	0.713	Transition rates
k_0	Vacancy cost: non-college jobs	35.90	Transition rates
k_1	Vacancy cost: humanities jobs	22.63	Transition rates
k_2	Vacancy cost: business jobs	169.48	Transition rates
k_3	Vacancy cost: STEM jobs	164.00	Transition rates
$c_{s,1}$	Switching cost: humanities graduates	-0.020	Transition rates
$c_{s,2}$	Switching cost: business graduates	0.230	Transition rates
$c_{s,3}$	Switching cost: STEM graduates	0.276	Transition rates
$\bar{a}^{mismatched}$	Prob. good match: mismatched workers	0.006	Transition rates

Note: This table reports all model parameters, grouped by calibration step. Panel A lists parameters that are normalized or assigned from external sources. Panel B reports the 14 production-function parameters calibrated by matching age-specific log wage profiles from the NLSY79 (Section 2.8). Panel C reports the 10 search-and-matching parameters calibrated by targeting age-specific UE, EU, and EE transition rates (Table 1).

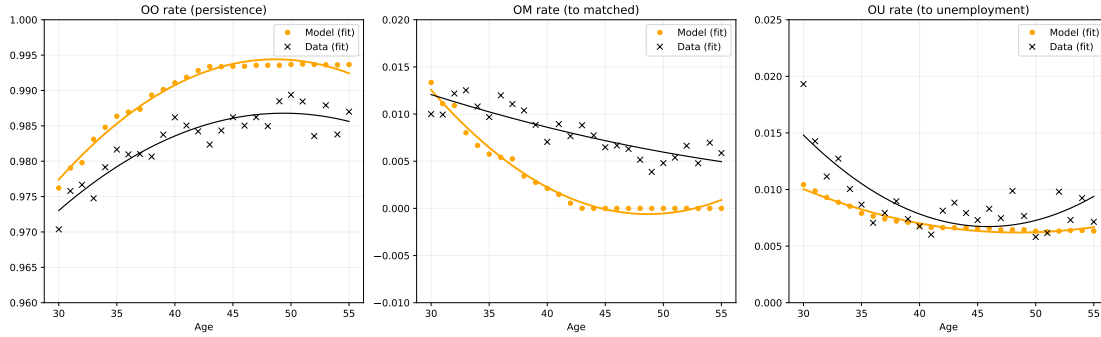
4.2 Validation: Overeducation Persistence in the Model

As a form of validation, I assess whether the calibrated model can replicate key features of overeducation dynamics that were *not* directly targeted in the calibration. The left-panel of Figure 10 provides the central comparison, showing the monthly overeducation

persistence rate (the share of overeducated workers who remain overeducated one month later) from both the calibrated model and the NLSY79 data. The model successfully replicates the empirical pattern: a high persistence rate throughout the lifecycle, rising from approximately 97% at age 30 to over 99% after age 40. Since nothing directly related to the overeducation persistence has been used in the calibration, the fact that both the magnitude and the lifecycle pattern are closely aligned with the data should be seen as an important result for the theory.

Figure 10 further decomposes the model's predictions for the transitions *out of* overeducation and compares them with the data. The overeducation-to-matched (OM) rate and the overeducation-to-unemployment (OU) rate are both untargeted moments. The model captures the qualitative patterns: both rates decline with age, consistent with the increasing persistence documented above. The OM rate in the model is somewhat lower than in the data, reflecting the fact that the empirical OM rate likely captures mechanisms beyond the scope of the model, such as within-field promotions or reclassifications that change the formal education requirement of the job. The OU rate is well matched by the model.

Figure 10: Untargeted overeducation transitions: Model vs. Data.



Note: This figure compares transition rates from overeducation in the calibrated model and the NLSY79 data (ages 30–55). Left panel: overeducation persistence (OO rate), i.e. the monthly probability that an overeducated worker remains overeducated. Center panel: overeducation-to-matched (OM) rate, i.e. the monthly probability of transitioning from an overeducated job to a matched job. Right panel: overeducation-to-unemployment (OU) rate, i.e. the monthly probability of moving from overeducation to unemployment. The x-axis reports worker age; the y-axis shows the monthly transition probability. Dots represent raw age-specific rates from the NLSY79; lines are polynomial fits to the model-implied rates. None of these moments is directly targeted in the calibration; they serve as out-of-sample validation. See Section 2.1 for sample restrictions and Section 2.2 for the overeducation/matched definition.

5 Decomposition of Overeducation Persistence

5.1 Decomposition Methodology

This section is dedicated to the key objective of this paper: decomposing the monthly persistence of overeducation into the main structural channels of the model over the lifecycle. To do so, I perform a counterfactual analysis. I compute the model-implied overeducation persistence rate (the OO rate) for ages 30 to 55 under the baseline calibration. I then sequentially shut down specific frictions in the model and re-simulate the OO rate. The difference between the baseline OO rate and a counterfactual OO rate represents the contribution of that specific friction to overeducation persistence.

I consider four distinct counterfactual scenarios:

1. **No Search Frictions:** I set the vacancy creation costs k_{f_j} to zero. This effectively eliminates search frictions and set the probability of finding a job for the worker to 1 (i.e., $p(\theta) = 1$). Notice that the switching costs c_{s,f_i} are still in place, to take into account only the search frictions explicitly modeled.
2. **No Experience Loss (Specialization Trap):** I assume that field-specific experience is perfectly retained when a worker switches to a job in a different field. In the baseline model, experience resets to zero upon changing fields. By removing this penalty, I shut down the specialization trap.
3. **No Learning Frictions:** I assume that match quality a_{ij} is perfectly known upon meeting (i.e., $\alpha = 1$). This eliminates the option value of staying in an overeducated job merely to learn about match quality or because of the uncertainty about alternative jobs.
4. **No Aging:** I assume that workers optimize as if they are forever 30 years old, shutting down the horizon effect (the approaching retirement) and skill depreciation.

5.2 Decomposition Results

Figure 11 presents the results of this counterfactual analysis. The solid blue line represents the baseline overeducation persistence rate, which is the same persistence computed with the calibrated model (See Figure 10).

The first result is that search frictions are the most important driver of overeducation persistence. Early in the career, absent any search frictions, overeducation persistence would fall to 50–60% (compared to 97% in the baseline). This is a huge drop: out of 1 million overeducated workers, 400,000 would exit overeducation one month later

(compared with only 25,000 in the baseline). As workers age, the importance of search frictions declines, but they remain the most important driver of persistence even at the end of the career: after age 45, persistence without search frictions would be around 85% (compared with 99% in the baseline).

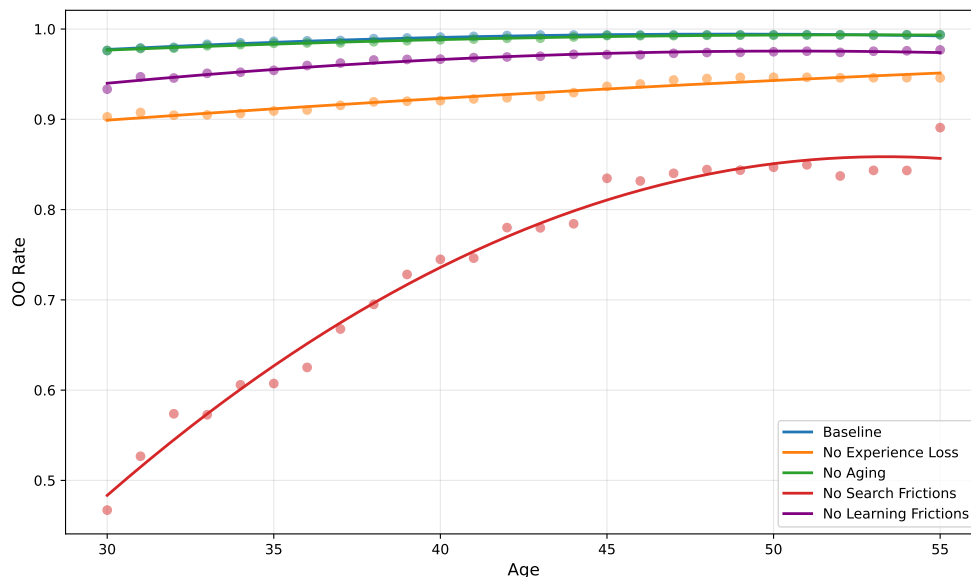
To better visualize the impact of the other channels, Figure 12 zooms in on the remaining counterfactuals by excluding the “No Search Frictions” scenario. The “No Experience Loss” scenario shows that the specialization trap is the second most important driver of overeducation persistence. When younger, if workers could retain their field-specific experience when switching fields, the persistence would be around 90%. Also in this counterfactual scenario, the importance of specialization decreases with age, and by the end of the career, overeducation persistence would fall to around 94%. Thus, the fact that workers accumulate non-transferable experience in their current field is a key factor that locks them into overeducated jobs, especially early in the career. However, the importance of this channel is much smaller than the role of search frictions. Learning frictions also contribute to persistence, although to a lesser extent. When match quality is known immediately, overeducation persistence would be around 94% at the beginning of the career and almost 98% at the end. Finally, aging has a minor role in overeducation persistence. The counterfactual scenario without aging shows that overeducation persistence would be the same as the baseline for the entire lifecycle. Thus, skill depreciation and the approaching retirement do not appear to be important drivers of overeducation persistence.

6 Conclusion

This paper has examined the persistence of overeducation over the lifecycle through the lens of a directed search model in which workers differ in educational level and field of study, field-specific experience, and age. Using U.S. data from the NLSY79 and educational requirements from O*NET, the model successfully replicates the empirical patterns of overeducation persistence without directly targeting them in calibration.

Four counterfactual scenarios isolate the contribution of each channel. First, search frictions are by far the most important driver: removing them would reduce monthly persistence from 97% to 50–60% early in the career, and from 99% to around 85% after age 45. Second, the specialization trap—driven by the accumulation of non-transferable field-specific experience—is the second most important channel, lowering persistence to about 90% when young and 94% later. Third, learning frictions contribute to a smaller but non-negligible extent, with persistence falling to around 94% early on when match quality is immediately known. Finally, aging effects have virtually no impact on persistence throughout the lifecycle.

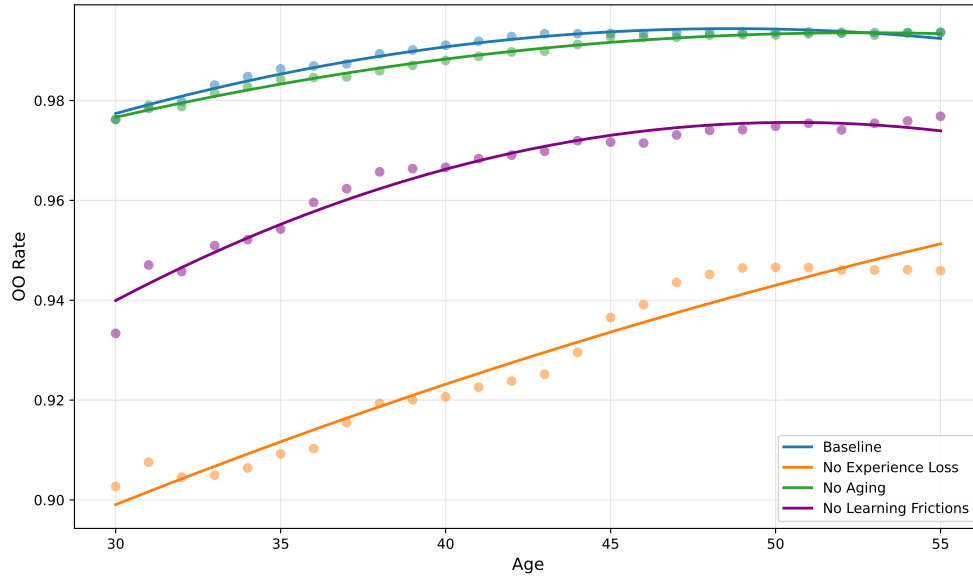
Figure 11: Decomposition of Overeducation Persistence: All Channels



Note: This figure plots the model-implied monthly overeducation persistence rate (the probability that an overeducated worker remains overeducated one month later) over the lifecycle (ages 30–55). The x-axis reports worker age; the y-axis shows the monthly persistence probability. The solid line is the baseline calibration. Four counterfactual scenarios are shown: (i) *No Search Frictions*—vacancy costs are set to zero so that all searching workers find a job with certainty; (ii) *No Experience Loss*—field-specific experience is retained when switching fields, removing the specialization trap; (iii) *No Learning Frictions*—match quality is immediately revealed ($\alpha = 1$), eliminating the option value of staying to learn; and (iv) *No Aging*—workers optimize as if they are forever 30, removing horizon and depreciation effects. See Section 5.1 for the decomposition methodology.

These findings highlight that overeducation is not merely a transient phenomenon but instead results from structural mechanisms that lock workers into mismatched jobs. Policy implications follow directly: because search frictions are the dominant channel, policies that reduce the cost of matching—particularly early in the career—can materially lower long-run overeducation. In addition, incentives that reduce the cost of switching fields help undo specialization-driven traps. Overall, a lifecycle perspective is essential for understanding and addressing the persistence of overeducation.

Figure 12: Decomposition of Overeducation Persistence: Role of Specialization, Learning, and Aging



Note: This figure reproduces Figure 11 but excludes the “No Search Frictions” scenario to better visualize the remaining channels. The x-axis reports worker age (30–55); the y-axis shows the monthly overeducation persistence probability. Three counterfactual scenarios are displayed alongside the baseline: (i) *No Experience Loss*—field-specific experience is fully retained when switching fields, removing the specialization trap; (ii) *No Learning Frictions*—match quality is immediately revealed ($\alpha = 1$); and (iii) *No Aging*—workers optimize as if they are forever 30. See Section 5.1 for the decomposition methodology and Figure 11 for the full comparison including search frictions.

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Appendix

A Proof of Block Recursive Equilibrium

The proof adapts the argument in Theorem 1 of Menzio et al. (2016) to accommodate the additional heterogeneity introduced in this paper: field-specific vacancy costs $k_{f'_j}$, proportional switching costs c_{s,f_i} , non-transferable field-specific experience, and field-dependent match quality priors. The strategy is backward induction on the finite horizon $t \in \{0, 1, \dots, T\}$.

Notation. Recall from Section 3.3 that a worker's type is $(\omega, t) = (a_{ij}, f_i, f_j, e_{if}, t)$. When a worker searches for a job in field f'_j , the new match state is $\omega' = (a^0, f_i, f'_j, e', t)$, where

$$e' = \begin{cases} e_{if} & \text{if } f'_j = f_j, \\ 0 & \text{if } f'_j \neq f_j. \end{cases}$$

As defined in the main text, the effective value of a new match, inclusive of the switching cost, is

$$\tilde{V}_t(\omega') = V_t(\omega') \times \begin{cases} 1 & \text{if } f'_j = f_j, \\ 1 - c_{s,f_i} & \text{if } f'_j \neq f_j. \end{cases}$$

Step 1: Terminal period T . In the last period, the value of unemployment is

$$U_T(\omega, \psi) = b(a^H, f_i, f_j, e_{if}, T),$$

which depends only on the worker's individual characteristics (through her most recent job), not on the aggregate state ψ . The joint surplus of a match is

$$V_T(\omega, \psi) = \begin{cases} y(a_{ij}, f_i, f_j, e_{if}, T) & \text{if } a_{ij} \in \{a^G, a^B\}, \\ \mathbb{E}[y(a_{ij}, f_i, f_j, e_{if}, T)] & \text{if } a_{ij} = a^0, \end{cases}$$

where the expectations are taken over match quality:

$$\mathbb{E}[y(a_{ij}, f_i, f_j, e_{if}, T)] = P(a_{ij}=a^G) y(a^G, f_i, f_j, e_{if}, T),$$

using $y(a^B, \cdot) = 0$. The probability $P(a_{ij}=a^G)$ depends on the education-job match as specified in Section 3.2. Since all terms on the right-hand side depend only on ω , both U_T and V_T are independent of ψ :

$$U_T(\omega, \psi) = U_T(\omega), \quad V_T(\omega, \psi) = V_T(\omega).$$

Since V_T does not depend on ψ , the effective value $\tilde{V}_T(\omega')$ is also independent of ψ . The zero-profit condition for a vacancy in the submarket targeting field f'_j is

$$k_{f'_j} \geq q(\theta_T(\omega', x)) [\tilde{V}_T(\omega') - x], \quad \theta_T(\omega', x) \geq 0,$$

with complementary slackness. Inverting on active submarkets:

$$\theta_T(\omega', x) = \begin{cases} q^{-1} \left(\frac{k_{f'_j}}{\tilde{V}_T(\omega') - x} \right) & \text{if } x \leq \tilde{V}_T(\omega') - k_{f'_j}, \\ 0 & \text{otherwise.} \end{cases}$$

Since $\tilde{V}_T(\omega')$ does not depend on ψ , neither does $\theta_T(\omega', x)$, i.e. $\theta_T(\omega', x, \psi) = \theta_T(\omega', x)$.

Search problem of the unemployed at T . An unemployed worker of type ω solves

$$R_T(\omega, \psi) = \max_{x, f'_j} \left\{ p(\theta_T(\omega', x)) [x - U_T(\omega)] \right\}.$$

On active submarkets, from the zero-profit condition, $x = \tilde{V}_T(\omega') - k_{f'_j}/q(\theta_T(\omega', x))$. Substituting:

$$R_T(\omega) = \max_{f'_j} \max_{\theta \geq 0} \left\{ -k_{f'_j} \theta + p(\theta) [\tilde{V}_T(\omega') - U_T(\omega)] \right\},$$

using $p(\theta)/q(\theta) = \theta$. Since x enters only through θ and all terms are ψ -independent, we have $R_T(\omega, \psi) = R_T(\omega)$.

For each candidate field f'_j , the inner maximization over θ has a unique solution $\theta_{f'_j}^*$ satisfying the first-order condition $p'(\theta_{f'_j}^*) = k_{f'_j}/[\tilde{V}_T(\omega') - U_T(\omega)]$ whenever the surplus $\tilde{V}_T(\omega') - U_T(\omega) > 0$ (and $\theta_{f'_j}^* = 0$ otherwise). The optimal field choice is

$$f_j^{*,u} = \arg \max_{f'_j} \left\{ -k_{f'_j} \theta_{f'_j}^* + p(\theta_{f'_j}^*) [\tilde{V}_T(\omega') - U_T(\omega)] \right\}.$$

The optimal market tightness and offered value are:

$$\theta_T^u(\omega) = \theta_{f_j^*, u}^*, \quad x_T^u(\omega) = \tilde{V}_T(\omega^*) - \frac{k_{f_j^*, u}}{q(\theta_T^u(\omega))},$$

where ω^* denotes the new match state at the optimally chosen field.

Search problem of the employed at T . An employed worker of type $(\omega(a), T)$ with known match quality $a \neq a^0$ who searches on the job solves

$$S_T(\omega(a), \psi) = \max_{x, f_j'} \left\{ p(\theta_T(\omega', x)) [x - V_T(\omega(a))] \right\}.$$

Here the outside option of not accepting the new offer is $V_T(\omega(a))$, the continuation value of the current match. Following the same substitution as for the unemployed:

$$S_T(\omega(a)) = \max_{f_j'} \max_{\theta \geq 0} \left\{ -k_{f_j'} \theta + p(\theta) [\tilde{V}_T(\omega') - V_T(\omega(a))] \right\}.$$

By the same argument, $S_T(\omega(a), \psi) = S_T(\omega(a))$. The optimal tightness $\theta_T^e(\omega(a))$ and offered value $x_T^e(\omega(a))$ are determined analogously to the unemployed case, with $V_T(\omega(a))$ replacing $U_T(\omega)$.

Separation policy at T . For a match with known quality a , the employment policy function d_T solves

$$\max_{d \in [\delta, 1]} \left\{ d U_T(\omega) + (1 - d) [V_T(\omega(a)) + \lambda_e S_T(\omega(a))] \right\}.$$

Since this is linear in d : $d_T = \delta$ if $V_T(\omega(a)) + \lambda_e S_T(\omega(a)) \geq U_T(\omega)$, and $d_T = 1$ otherwise. All terms are ψ -independent, so $d_T(\omega, \psi) = d_T(\omega)$.

Summary at T . All value functions (U_T, V_T), search values (R_T, S_T), policy functions ($\theta_T^u, \theta_T^e, d_T$), and market tightness (θ_T) depend on ω but not on the aggregate state ψ .

Step 2: Induction from $t + 1$ to t . Suppose that at $t + 1$, all value and policy functions are independent of ψ , that is, $U_{t+1}(\omega, \psi) = U_{t+1}(\omega)$, $V_{t+1}(\omega, \psi) = V_{t+1}(\omega)$, $R_{t+1}(\omega, \psi) = R_{t+1}(\omega)$, $S_{t+1}(\omega, \psi) = S_{t+1}(\omega)$, and $\theta_{t+1}(\omega', x, \psi) = \theta_{t+1}(\omega', x)$. We show this implies the same at t .

Value of unemployment at t . From Section 3.5:

$$U_t(\omega, \psi) = b(a^H, f_i, f_j, e_{if}, t) + \beta \mathbb{E}_{\hat{\psi}|\psi} [U_{t+1}(\omega, \hat{\psi}) + \lambda_u R_{t+1}(\omega, \hat{\psi})].$$

By the induction hypothesis, $U_{t+1}(\boldsymbol{\omega}, \hat{\psi}) = U_{t+1}(\boldsymbol{\omega})$ and $R_{t+1}(\boldsymbol{\omega}, \hat{\psi}) = R_{t+1}(\boldsymbol{\omega})$. Therefore, the expectation $\mathbb{E}_{\hat{\psi}|\psi}[\cdot]$ operates on terms that do not depend on $\hat{\psi}$, so it drops out:

$$U_t(\boldsymbol{\omega}) = b(a^H, f_i, f_j, e_{if}, t) + \beta [U_{t+1}(\boldsymbol{\omega}) + \lambda_u R_{t+1}(\boldsymbol{\omega})].$$

Hence $U_t(\boldsymbol{\omega}, \psi) = U_t(\boldsymbol{\omega})$.

Joint match value at t (known quality). For $a \in \{a^G, a^B\}$, using the fact that $\hat{e}_{if} = e_{if} + 1$:

$$\begin{aligned} V_t(\boldsymbol{\omega}(a), \psi) &= y(a, f_i, f_j, e_{if}, t) + \beta \mathbb{E}_{\hat{\psi}|\psi} \left[\max_{d \in [\delta, 1]} \{d U_{t+1}(\hat{\boldsymbol{\omega}}(a)) + \right. \\ &\quad \left. + (1-d)(V_{t+1}(\hat{\boldsymbol{\omega}}(a)) + \lambda_e S_{t+1}(\hat{\boldsymbol{\omega}}(a)))\} \right]. \end{aligned}$$

By the induction hypothesis, U_{t+1} , V_{t+1} , and S_{t+1} are all independent of $\hat{\psi}$. Therefore the max operator inside the expectation does not depend on $\hat{\psi}$, and the expectation drops out. Hence $V_t(\boldsymbol{\omega}(a), \psi) = V_t(\boldsymbol{\omega}(a))$.

Joint match value at t (unknown quality). For $a_{ij} = a^0$:

$$\begin{aligned} V_t(\boldsymbol{\omega}(a^0), \psi) &= \alpha \mathbb{E}[V_t(\boldsymbol{\omega}(a), \psi)] + (1-\alpha) \mathbb{E}[y(a_{ij}, f_i, f_j, e_{if}, t)] + \\ &\quad + (1-\alpha) \beta \mathbb{E}_{\hat{\psi}|\psi} \left[\max_{d \in [\delta, 1]} \{d U_{t+1}(\hat{\boldsymbol{\omega}}) + \right. \\ &\quad \left. + (1-d)(V_{t+1}(\hat{\boldsymbol{\omega}}(a^0)) + \lambda_e S_{t+1}(\hat{\boldsymbol{\omega}}(a^0)))\} \right]. \end{aligned}$$

The first term involves $V_t(\boldsymbol{\omega}(a^G))$ and $V_t(\boldsymbol{\omega}(a^B))$, which we have just shown to be ψ -independent at t . The expected output $\mathbb{E}[y]$ depends only on $\boldsymbol{\omega}$. For the continuation, the induction hypothesis implies that U_{t+1} , V_{t+1} , and S_{t+1} at a^0 are $\hat{\psi}$ -independent, so the expectation $\mathbb{E}_{\hat{\psi}|\psi}$ drops out. Hence $V_t(\boldsymbol{\omega}(a^0), \psi) = V_t(\boldsymbol{\omega}(a^0))$.

Effective match value and market tightness at t . Since V_{t+1} is ψ -independent, $\tilde{V}_{t+1}(\boldsymbol{\omega}')$ is ψ -independent. The zero-profit condition at $t+1$,

$$k_{f'_j} \geq q(\theta_{t+1}(\boldsymbol{\omega}', x)) [\tilde{V}_{t+1}(\boldsymbol{\omega}') - x], \quad \theta_{t+1} \geq 0,$$

involves only ψ -independent terms, confirming the induction hypothesis on θ_{t+1} .

Search values at t . The unemployed search value becomes

$$R_{t+1}(\boldsymbol{\omega}) = \max_{f'_j} \max_{\theta \geq 0} \left\{ -k_{f'_j} \theta + p(\theta) [\tilde{V}_{t+1}(\boldsymbol{\omega}') - U_{t+1}(\boldsymbol{\omega})] \right\},$$

which is ψ -independent. Analogously, the employed search value is

$$S_{t+1}(\hat{\omega}(a)) = \max_{f'_j} \max_{\theta \geq 0} \left\{ -k_{f'_j} \theta + p(\theta) [\tilde{V}_{t+1}(\omega') - V_{t+1}(\hat{\omega}(a))] \right\},$$

which is also ψ -independent. In both expressions, $\omega' = (a^0, f_i, f'_j, e', t+1)$ with $e' = e_{if} \mathbf{1}[f'_j = f_j]$, and the effective value $\tilde{V}_{t+1}$ incorporates the switching cost c_{s,f_i} when $f'_j \neq f_j$.

Separation policy at t . The linearity of the $\max_d$ problem in d , together with the ψ -independence of U_{t+1} , V_{t+1} , and S_{t+1} , yields $d_t(\omega, \psi) = d_t(\omega)$ by the same argument as at T .

Conclusion. By induction, all value functions, policy functions, and market tightness are independent of the aggregate state ψ at every t . This establishes the block-recursive structure of the equilibrium. ■